

Concept Question 5.1 The Sun in a cluster

Would the Sun have been more likely to be a member of a globular or open cluster in the past?

Problem 5.2 Close encounters

In the lecture we treat stars as point masses and estimate the velocity deflection during two-body interaction in a distant encounter.

- a) Recall the criterion of the impact parameter b for a distant encounter.
- b) The criterion is not satisfied if the two stars encounter at a closer distance. In a general situation, the encounters are described by unbound Keplerian orbits. Assuming two point masses m_1 and m_2 at position $\vec{x}_1$ and $\vec{x}_2$ with velocities $\vec{v}_1$ and $\vec{v}_2$

1. Prove that the motions obey the equation:

$$\mu \ddot{\vec{r}} = -\frac{G(m_1 + m_2)\mu}{r^2} \vec{e}_r \quad (1)$$

where $\vec{r} = \vec{x}_2 - \vec{x}_1$, $\mu = \frac{m_1 m_2}{m_1 + m_2}$.

2. Use the solution of motion m in a Kepler potential of the fixed body M :

$$r(\psi)^{-1} = C \cos(\psi - \psi_0) + GM/L^2 \quad (2)$$

where L being the conserved angular momentum per unit mass associated with m . Determine the ψ_0 using initial condition at $t = -\infty$ ($r = -\infty$ and $v = -v_0$).

3. The impact parameter b_{90} is defined as the deflection radius at which the relative velocity is deflected by 90° . Derive

$$b_{90} \equiv G(m_1 + m_2)/v_0^2 \quad (3)$$

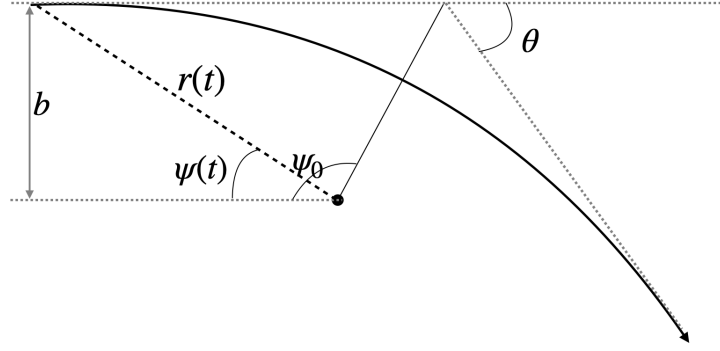


Figure 1: The motion of the reduced particle during hyperbolic encounter

- c) As shown in **b)**, a binary cannot form from the isolated two-body encounter. However a close encounter involving 3 stars can leave two participants in a bound Keplerian orbit. Let's consider a triple encounter, when 3 stars approach each other within distance b_{90}
1. What is the characteristic timescale for a star to experience a encounter closer than b_{90} ? Assuming the number density of stars $n \approx N/r^3$, close encounter cross section $\Sigma \approx b_{90}^2$, $v^2 \approx GNm/r$.
 2. What is the probability of a third star lying within b_{90} at each encounter? Derive the characteristic timescale t_3 for a triple encounter less than b_{90} . Use the approximation $b_{90} \approx Gm/v^2$. Compare this timescale with the relaxation time scale. Is triple encounter the main origin of binary formation in globular clusters?